

👤 Professional Profile

AI & ML Engineer at **Solver Intelligent Analytics**, where I joined the team after completing my internship (Jan–Jul 2026, 560 h). I design and ship AI systems to production for real clients: the conversational RAG assistant for the Comú d'Encamp (Andorra), with administrative document generation built on a **LangGraph** pipeline and **python-docx** on top of a self-hosted vLLM instance, and **PPTX Updater**, a document automation tool in production at Food Delivery Brands (Telepizza).

Final-year Computer Science student specialized in **AI systems**, **advanced RAG** and **backend development**. Hands-on experience building **RAG pipelines** (HyDE, MultiQuery, rerankers, Chroma Cloud), **agents** with LangChain/LangGraph/CrewAI/AutoGen and **Transformer** and **CNN** models from scratch. Designed multi-agent debate systems, implemented a *Vision Transformer*, and developed scalable web applications. Advocate of **Scrum**, with experience as developer and Scrum Master, delivering robust and well-documented solutions.

🎓 Education

Universitat de València

Valencia, Spain

Bachelor's Degree in Computer Science Engineering

Sep 2022 – Sep 2026 (expected)

Key subjects: Data structures and algorithms, Web application development, Intelligent systems, Software engineering, Databases.

💼 Experience

Solver Intelligent Analytics

Valencia, Spain

AI & ML Engineer, Hybrid

Jul 2026 – Present

Joined the AI team after the internship. Building and evolving production AI systems for clients: multilingual conversational RAG assistants, agents on self-hosted LLMs, and document automation (Word, PowerPoint, Excel) integrated with SharePoint and Microsoft Graph.

Stack: Python, LangGraph, FastAPI, vLLM, Docker, React, Microsoft Graph, SharePoint, xlwings/COM, python-pptx.

Solver Intelligent Analytics

Valencia, Spain

Machine Learning Engineer, Internship contract · Hybrid · 7 months · 560 h

Jan 2026 – Jul 2026

Built a conversational RAG assistant for the Comú d'Encamp (Andorra): a multilingual production chatbot integrating multiple functionalities for the municipal administration. One of the main features is the automatic generation of public procurement documents: through a guided conversational flow orchestrated with **LangGraph**, the system collects the required information from the user and produces the final Word document, covering twelve variants depending on the procedure type. The module covers the end-to-end flow: data extraction and validation, XML-level DOCX templating, and integration with **SharePoint** via **Microsoft Graph** to keep the client's templates and examples synchronized.

Stack: Python, LangGraph, python-docx, FastAPI, Docker, vLLM (Qwen3-30B), Microsoft Graph, SharePoint, RAG, prompt engineering.

💻 Projects

📦: **PPTX Updater** 🚀 in production · Food Delivery Brands (Telepizza)

Python, FastAPI, xlwings/COM (win32api), python-pptx, openpyxl, React, MSAL, Microsoft Graph, SharePoint, PyInstaller, Inno Setup, pytest, Playwright

Desktop automation tool used in production by Food Delivery Brands (Telepizza) to regenerate monthly board

decks from Excel: the tool scans the PowerPoint, lets the user map each chart/table to a workbook, sheet and cell range, captures those ranges as images through a background Excel COM engine (direct chart export + vector rendering, validation and retry layers, never stealing the user's focus), and replaces the shapes in place at natural scale. Shape identity survives month-over-month reuse (UUID markers embedded in the deck), so a generated presentation can serve as next month's template. Includes a FastAPI backend with a single-page React UI, SharePoint integration via Microsoft Graph with Entra ID sign-in (multi-site/multi-library projects, sharing and role-based access) deployed live on the client's tenant, and delivery as a self-contained Windows installer (PyInstaller + Inno Setup) requiring no Python or admin rights.

: **yolo_v1**  [repo](#)

[PyTorch](#), [DDP](#), [AMP](#), [COCO 2017](#), [ResNet18](#)

From-scratch implementation of **YOLOv1** (Redmon et al., 2016) trained on COCO 2017 with 80 classes. Two backbones (Darknet from scratch and pretrained ResNet18), 7×7 detection grid with 2 boxes per cell, the original five-term loss and distributed training (DDP) with mixed precision (AMP); ships 38 pytest tests and a detailed write-up of silent bugs.

: **asistente-uv** (Bachelor's Thesis)  [repo](#)

[LangGraph](#), [Ollama](#), [ChromaDB](#), [BM25](#), [Playwright](#), [FastAPI](#), [React](#), [Docker](#)

RAG assistant that answers questions about the University of Valencia using official information extracted from [uv.es](#). Combines a local document store with real-time web search when information is missing, and grows automatically with each new query. ReAct-style agent on LangGraph with hybrid retrieval (BM25 + dense + RRF) and Playwright-based scraping.

: **CodeScribe**  [repo](#)

[Python](#), [LangGraph](#), [Ollama](#), [Click](#), [Rich](#), [Pydantic](#), [uv](#)

CLI tool that scans an entire project and adds documentation and comments to every source file using a local LLM via Ollama. Deterministic LangGraph pipeline: scan → classify → chunk → LLM → post-process → write back. Supports dry-run with diff, backups, interactive menu and automatic README generation.

: **chatbot — LangGraph RAG Assistant**  [repo](#)

[LangGraph](#), [BM25](#), [Dense Embeddings](#), [RRF](#), [SQLite](#), [Selenium](#)

Conversational QA system with hybrid retrieval fused via Reciprocal Rank Fusion, strict chunk-level citations, adaptive Selenium-based web enrichment and short/long-term memory backed by SQLite. Orchestrated in a single LangGraph instance with clearly contracted nodes. End-to-end multilingual.

: **llama_4_scratch**  [repo](#)

[PyTorch](#), [NumPy](#), [Jupyter](#)

From-scratch implementation and step-by-step explanation of the key components of a LLaMA-style **decoder Transformer**: tokenizer, scaled dot-product attention and Multi-Head Latent Attention. Every operation commented with numerical examples and visualizations.

: **multiquery-hyde-rag-chroma-cloud**  [repo](#)

[Python](#), [RAG](#), [LangChain](#), [Chroma Cloud](#)

Advanced RAG pipeline with **HyDE** and **MultiQuery** to expand coverage, indexing in **Chroma Cloud**, and reranking to improve response accuracy; includes retrieval evaluation and reproducible documentation.

: **advanced-hyde-rag**  [repo](#)

[Python](#), [RAG](#), [LangChain](#)

RAG implementation with HyDE and synthetic query generation, chunking and embeddings handling, comparison of retrieval strategies and latency analysis.

: **ranked-rag-pipeline**  [repo](#)

[Python](#), [Cross-Encoder](#), [Sentence Transformers](#)

Retrieval chain with **rerankers** and filters, coverage/precision metrics and utilities for source traceability.

: **bm25-dense-retrieval**  [repo](#)

[BM25](#), [Embeddings](#), [RRF](#)

Practical study of hybrid retrieval combining BM25 (lexical) with dense embeddings (semantic) and Reciprocal Rank Fusion. Performance comparison across query types.

: **data-chat** [repo](#)

[LangChain](#), [LLMs](#), [SQL](#)

Exploration of **text-to-SQL** and data chat patterns with agents: translate natural language questions into structured queries, execute them and return summarized answers.

: **MultiagentDebate** [repo](#)

[Python](#), [LangChain/LangGraph](#)

Multi-agent debate system (scientific, economic, historical, psychological, refutation) with supervisors and automatic evaluation; orchestration with LangGraph and evidence logging.

: **VisionTransformer** [repo](#)

[PyTorch](#), [Computer Vision](#)

Educational implementation of a Vision Transformer (ViT): patch embedding, multi-head attention, MLP head and training on vision datasets, with comparative analysis against CNN architectures.

: **PoetryTransformerPredict** [repo](#)

[PyTorch](#), [NLP](#)

Bigram Transformer built from scratch (~9.5k embedding, 4 attention blocks) to generate Spanish poetry; training and sampling example.

: **GiveMeSomeCredit — Default Risk** [repo](#)

[PyTorch](#), [scikit-learn](#), [pandas](#), [MLP](#), [AMP](#)

Clean, reproducible baseline to predict serious delinquency within two years with PyTorch. Data audit, median imputation, quantile clipping, log1p transforms, RobustScaler, MLP with BatchNorm and GELU, mixed-precision training and early stopping on ROC-AUC.

: **Recommender Systems** [repo](#)

[Surprise](#), [NumPy](#), [Collaborative Filtering](#)

Collaborative filtering and content-based models, metrics (RMSE, MAP@K) and cross-validation; explanatory notebooks.

: **Brain Tumor Classification** [repo](#)

[TensorFlow](#), [VGG19](#)

Brain tumor vs. non-tumor classification through fine-tuning of VGG19 and data augmentation; validation set evaluation.

: **Amazigh Alphabets Recognition** [repo](#)

[TensorFlow](#), [Keras](#)

CNN network for tifinagh handwritten character recognition with > 95% accuracy; preprocessing and training pipeline.

: **Handwritten Alphabets & Digit Recognition** [repo](#)

[TensorFlow](#), [Keras](#), [MNIST](#)

CNN models for handwritten Latin letters and digits (MNIST), comparison with SVM/MLP and error analysis.

: **Housing Price Prediction** [repo](#)

[Python](#), [scikit-learn](#)

Linear regression and Random Forest for housing price estimation; RMSE and R^2 comparison.

: **TiendaWeb (Klawz)** [repo](#)

[Java/JSP](#), [MySQL](#)

Complete e-commerce with catalog, cart and payment gateway; MVC architecture, sessions and basic security.

: **PhongIllumination** [repo](#)

[C](#), [OpenGL](#), [OpenMP](#), [MPI](#)

Implementation of the Phong illumination model in sequential, shared-memory and distributed parallel versions; speedup measurement.

: **minilang** [repo](#)

[Python](#), [PLY](#)

Educational compiler for MiniLang with lexical, syntactic and semantic analysis, AST generation and test execution.

Certifications & Courses

Specializations & Programs

Nov 2025: PyTorch for Deep Learning (Professional Certificate) — DeepLearning.AI

ID: ZYEUD91L9FYS · Deep learning with PyTorch · Transfer learning for CV & NLP · Model optimization and deployment

Oct 2025: Building AI Agents and Agentic Workflows (Specialization) — IBM

ID: UROX1SUIDGWA · Design of agentic workflows, tool use, orchestration and evaluation

Oct 2025: AI for Medicine Specialization — DeepLearning.AI

ID: 6MOISRA2N5WK · Precision medicine, image analysis, risk modeling, decision trees, forecasting

Sep 2025: RAG for Generative AI Applications (Specialization) — IBM

ID: W2UQZCZMJHKV · End-to-end RAG system design and deployment

Feb 2025: Deep Learning Specialization — DeepLearning.AI

ID: EOAJ4XETDWMR · Neural networks, CNN, optimization, regularization, sequence models

Nov 2024: Machine Learning — DeepLearning.AI (Andrew Ng)

ID: 9PL5C8QDGYQ6 · Supervised, unsupervised, regularization, evaluation

Nov 2024: IBM Introduction to Machine Learning — IBM

ID: 59X2YJWXX9SS · Foundations of ML, supervised and unsupervised learning, model evaluation

RAG & Agents

Sep 2025: Advanced RAG with Vector Databases and Retrievers — IBM

ID: TSXE4ZG7MDFQ

Sep 2025: Agentic AI with LangGraph, CrewAI, AutoGen and BeeAI — IBM

ID: AWXAVIH36K1F

Sep 2025: IBM RAG and Agentic AI — IBM

ID: X3774PPHATF6

Jul 2025: Build RAG Applications: Get Started — IBM

ID: 596YXNIVML4A

Jul 2025: Fundamentals of Building AI Agents — IBM

ID: ZGQWDFK0VO5H

Jul 2025: Vector Databases for RAG: An Introduction — IBM

ID: K46CQ74O5CKA

Jul 2025: AI Enhancement with Knowledge Graphs — Mastering RAG Systems — Packt

ID: 28DS7P6LJIE8

Jul 2025: Agentic AI with LangChain and LangGraph — IBM

ID: V82LDS06UQN4

Jul 2025: Develop Generative AI Applications: Get Started — IBM

ID: 2EPBGK0YLTIP

Jul 2025: Build Multimodal Generative AI Applications — IBM

ID: IW6XFBO3EGSN

PyTorch & Deep Learning

Nov 2025: PyTorch: Fundamentals — DeepLearning.AI

ID: 6PJ2XY48LZZO · PyTorch basics, Tensors and Autograd

Nov 2025: PyTorch: Techniques and Ecosystem Tools — DeepLearning.AI

ID: CR0RUHDBQ5CS · Torch.Optim, PyTorch Lightning, ecosystem tools

Nov 2025: PyTorch: Advanced Architectures and Deployment — DeepLearning.AI

ID: CK4L1IG1BJ5L · Encoder–Decoder, ResNet, DenseNet, deployment

Sep 2025: Building and Training Neural Networks with PyTorch — Packt

ID: N4BCRQSYJ7D2 · CNNs

Sep 2025: Foundations and Core Concepts of PyTorch — Packt

ID: OJG0HNZ0ZQC9 · ML fundamentals

 NLP & Transformers

Mar 2025: Transformer Models and BERT Model — Google Cloud

ID: K2LDLD7XJIM6

Feb 2025: Attention Mechanism — Google Cloud

ID: LJPL5IN4LHON

 ML/DL Fundamentals

2024–2025: Convolutional Neural Networks

ID: Y4QTHRNL3HE7 · TensorFlow, CNN, facial recognition, object detection & segmentation

2024–2025: Exploratory Data Analysis for Machine Learning

ID: 8PVON5MQJ62R · EDA, feature engineering, statistical tests

2024–2025: Improving Deep Neural Networks

ID: CNCPQJS4KTFE · Regularization, optimization, hyperparameters

2024–2025: Neural Networks and Deep Learning

ID: KNJNB8WQWAJP

2024–2025: Structuring Machine Learning Projects

ID: Z3KQ6DQXD2GM

2024–2025: Supervised Machine Learning: Classification

ID: HYIO24EKA6SX

2024–2025: Supervised Machine Learning: Regression

ID: LDQ28KIIHHG7

2024–2025: Unsupervised Machine Learning

ID: 52M2R676600S

Nov 2024: Supervised Machine Learning: Classification (Badge)

Digital certificate

 Web, Cloud & Networks

Feb 2025: Google AI Essentials — Google

ID: X51IOTBS2BSE

Feb 2025: Advanced Styling with Responsive Design — Univ. Michigan

ID: VHSLGPRZGP26

Feb 2025: Interactivity with JavaScript — Univ. Michigan

ID: JODZD9YLKDBT

Feb 2025: Introduction to CSS3 — Univ. Michigan

ID: FDU5K7VCNMBP

Feb 2025: Introduction to HTML5 — Univ. Michigan

ID: B9898ISXJUTR

Feb 2025: Introduction to Networking — NVIDIA

ID: AUQMCFVGVRP0

Feb 2025: AI Infrastructure and Operations Fundamentals — NVIDIA

ID: NFNQ1HP3ODGH

Technical Skills

Languages: Python, C/C++, Java, JavaScript, HTML5, CSS3, SQL

AI / ML: PyTorch, TensorFlow, Keras, scikit-learn, Transformers, BERT/ViT, LangChain, LangGraph, CrewAI, AutoGen, BeeAI, Ollama, vLLM

RAG / NLP: ChromaDB (Cloud), FAISS, BM25, HyDE, MultiQuery, Reranking, Retrieval Evaluation, Prompt Engineering, Knowledge Graphs

Backend: FastAPI, Flask, Java EE (JSP/Servlets), REST APIs, python-docx, python-pptx, openpyxl, xlwings/COM

Data: PostgreSQL, MySQL, SQLite, MongoDB, pandas, NumPy

DevOps: Docker, Docker Compose, Git & GitHub Actions, Google Cloud, Linux, PyInstaller, Inno Setup

Methodologies: Scrum (*developer and Scrum Master*)

Highlighted Skills

Advanced RAG (HyDE, MultiQuery, rerankers) · Vector databases (Chroma, FAISS) · Agents (LangGraph, CrewAI, AutoGen, BeeAI) · Transformers (BERT, ViT) · Convolutional networks · Risk modeling and forecasting · Decision trees · Image analysis · EDA & Feature engineering · Optimization & hyperparameters · PyTorch fundamentals · Document generation with templates

Soft Skills

Multicultural teamwork · Trilingual communication (ES/EN/FR) · Analytical thinking · Time management · Continuous learning · Academic leadership

Languages

Berber: Native

Arabic: Native

Spanish: C1–C2

Full professional proficiency

English: B2–C1

Full professional proficiency

French: B2

Professional working proficiency

Interests

Generative AI · Machine Learning · NLP · RAG · Databases · Compilers · Cloud Computing · Cybersecurity · Robotics · IoT · Video game development · Open Source · Optimization · Computer Vision · Drone photography · Hiking · Virtual/augmented reality